

Different factors control organic matter degradation in bulk and rhizosphere soil from the top- and subsoils of three forest stands

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ABSTRACT

The input of substrates and nutrients into deeper soil layers is limited to small regions of the bulk soil through living roots and preferential flow paths which can stimulate a more abundant and diverse microbial community. The aim of this study was to elucidate if SOC mineralization in bulk and rhizosphere soil is differentially limited by nutrients and substrates and to what extent this is site-specific. An 82-day incubation study was conducted with bulk and rhizosphere soil samples from the topsoil (0–10 cm) and subsoil (30–50 cm) from three beech forest sites (*Fagus sylvatica* L.) in northern Germany. All sites are within the same climatic region and share the same land use history and a similar old tree population but differ in parent material (Pleistocene sand = PS, Triassic sandstone = TS and Loess = LS). To identify the response of the microbial activity to changing nutritional conditions, ¹⁴C-substrate and SOC mineralization, microbial activity, enzyme activities from the C-, N-, and P-cycle, dehydrogenase and peroxidase activity were determined after substrate- and nutrient-additions. We found high amounts of labile C in the subsoil at the LS and PS sites and a generally higher SOC and faster substrate mineralization in the rhizosphere samples compared to the respective bulk soil in both, the topsoil (29–296%) and subsoil (73–563%) compared to the respective bulk soil. The addition of glucose caused positive priming effect only in samples from one site (TS), while effects were negative (PS) or not significant (LS) at the other sites. While glucose mineralization was similar in all samples, reaching 50–65% after 82 days, the mineralization of palmitic acid showed a differential pattern across the three sampling sites. In samples from two sites (PS, LS), the mineralization of palmitic acid was delayed by 10–30 days and had no effects on SOC turnover, while it showed no lag phase at the TS site and additionally stimulated SOC turnover. N additions generally and in some cases also P additions decreased SOC mineralization compared to the control samples. The results therefore show that labile and more complex substrates induce SOC mineralization differently between the investigated sample sites, with further differentiation with depth, sampling type and soil type, indicating that microorganisms in bulk and rhizosphere soil undergoes different adaptation processes based on their unique community composition.

1. Introduction

Subsoils contain large carbon pools as more than 50% of the soil organic carbon (SOC) is stored in deeper soil layers below 30 cm under the surface (Jobbagy and Jackson, 2000; Rumpel and Kögel-Knabner, 2011; O'Rourke et al., 2015). In the upper soil layers, fresh organic matter (OM) inputs from decaying litter and root turnover (Kuzyakov, 2002) provide the ideal environment for an abundant and active microbial community rapidly decomposing OM. With more distance to the surface however, SOC content and microbial biomass decline along the soil profile (Schöning and Kögel-Knabner, 2006; Schöning et al., 2006)

and the distribution of SOC becomes more heterogeneous with increasing depth (Angst et al., 2016a; Heinze et al., 2018). This is accompanied by an increase in the apparent ¹⁴C age (Jenkinson et al., 2008; Chabbi et al., 2009; Angst et al., 2016b; Herre et al., 2021) and significant changes in the microbial community composition (Fierer et al., 2003b; Preusser et al., 2017). In addition, the potential consumers are more separated from the sporadic and spatially limited substrate inputs (Heitkötter and Marschner, 2018a), leading microorganisms to be mainly in a dormant state (Schimel et al., 2007; Joergensen and Wichern, 2018; Heitkötter and Marschner, 2018b). As the input of labile C sources in subsoils is therefore limited to small volumes of the bulk

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soil, the formation of so called “hotspots” of microbial life (Fierer et al., 2003a; Kuzyakov and Blagodatskaya, 2015) is an important factor for regulating C turnover in deeper soil layers. These hotspots are often found along preferential flow paths (Bundt et al., 2001; Heinze et al., 2018) and in the rhizosphere soil, where a much more diverse and abundant microbial community thrives on constant influxes of DOC and exudates from the roots (Rasse et al., 2005; Rumpel and Kögel-Knabner, 2011; Angst et al., 2016b). Although most of the SOC in subsoils is stabilized for thousands of years (Rethemeyer et al., 2005; Kögel-Knabner et al., 2008; Angst et al., 2016a), incubation studies have shown that microorganisms in the bulk soil outside of hotspots can be triggered and “re-activated” when soil samples are homogenized, eliminating the spatial separation. The labile fraction of native OC can then be rapidly mineralized without additional substrate application (Salaomé et al., 2010; Wordell-Dietrich et al., 2017; Heitkötter et al., 2017b; Herre et al., 2021). To quantify changes in SOC decomposition after substrate addition, the term “priming effect” was established (Bingeman et al., 1953; Jenkinson, 1966; Kuzyakov et al., 2000), which describes a significant increase (*positive priming*) or decrease (*negative priming*) in SOC derived CO₂ emission (Blagodatskaya and Kuzyakov, 2008; Kuzyakov, 2010) compared to a control sample. So far, the reaction of microorganisms to different substrate additions is inconsistent (Hamer and Marschner, 2005; Salaomé et al., 2010; Guenet et al., 2010; Shahzad et al., 2019; Feng et al., 2021). When SOC decomposition is altered solely through the presence of living roots, the term “*rhizosphere priming effect*” (Zhu et al., 2014; Cheng et al., 2014) is applied. Through climate change, mean temperatures also rise in subsoils (Jia et al., 2019) causing plants to extend their root system even below 30 cm which can alter substrate and nutrient input in deeper soil layers (Wang et al., 2017b; Liu et al., 2018) and trigger the mineralization of previously stabilized OC. The emitted CO₂ will further contribute to rising greenhouse gas concentrations in the atmosphere (Soong et al., 2021). It is therefore necessary to further investigate C dynamics in subsoils to understand the mechanisms of the microbial turnover and help improving global carbon models (Wang et al., 2017a).

The aim of this study was to elucidate if SOC mineralization in bulk and rhizosphere soil of three beech forest sites developed on various parent material (Pleistocene sand, Triassic sandstone and Loess) is differentially limited by nutrients and substrates in the topsoil and subsoil (0–10 cm and 30–50 cm), where microbial community composition and OM input quality and quantity varies significantly (Herre et al., 2021). All three beech forest sites (*Fagus sylvatica* L.) are located within the same climatic region in northern Germany and share the same land use history and a similar old tree population making it possible to limit influencing factors to potential limitations due to different pedogenesis. To identify the response of the microbial community on changing nutritional conditions, microbial activity, enzyme activities from the C-, N-, and P-cycle, dehydrogenase and peroxidase activity after substrate- and nutrient-additions were measured. We hypothesized that (1) nutrient and substrate limitations are less pronounced in the rhizosphere soil due to continuous inputs from root exudates and root hair turnover; (2) nutrient and substrate limitations are more pronounced in the subsoil because of fewer input sources of fresh substrate; and (3) nutrient limitations are most pronounced in the sandy subsoils due to low OM content and cation exchange capacity.

2. Material and methods

2.1. Sampling sites

Soil samples were collected from three beech forest (*Fagus sylvatica* L.) sites in northern Germany differing in parent materials (Pleistocene sand = PS, Triassic sandstone = TS and Loess = LS) (Bundesanstalt für Bodenforschung, 1973) during a field campaign in April 2017. The soils are characterized as Dystric Cambisols (PS/TS site) and a Luvisol (LS site) (FAO, 2014). The sites were chosen because they have a similar old

tree population (120–130 y) of the most abundant and dominating tree species of the potential natural vegetation of central Europe (Ellenberg and Leuschner, 2010) and are within the same climatic region (Kirfel et al., 2019).

2.2. Soil sampling procedure

For each sampling site, three pits were excavated and soil samples were collected from two different depth increments. The topsoil and subsoil samples were taken from 0 to 10 cm and 30–50 cm, respectively. The bulk soil samples from each depth were collected in parts of the profile without roots being present. To sample the rhizosphere soil, root fragments with adhering soil were excavated and gently shaken by hand. The remaining soil adhering to the root was considered as rhizosphere and everything else was discarded (Yanai et al., 2003). Both bulk and rhizosphere soil were freshly sieved (<2 mm) and stored at 4 °C and –20 °C until analysis. To determine bulk density at each sample site and depth, undisturbed soil samples were taken additionally from each depth increment using a core sampler (100 cm³).

2.3. Physico-chemical soil analysis

Total nitrogen content (N_T) and soil organic carbon (SOC) were analyzed by dry combustion with a *Vario EL analyzer* unit (Elementar, Hanau, Germany). Samples were dried for three days at 50 °C and ground before analysis. The pH was measured in a 0.01 M CaCl₂ solution (ratio 1:2.5). Soil texture was determined with the laser particle sizer *Analysette 22 MicroTech plus* (FRITSCH GmbH, Idar-Oberstein, Germany). Samples with a SOC content >1.2% were pretreated with H₂O₂ to destroy organic matter in the soil.

2.4. Experimental design of the incubation

Soil samples were incubated for 82 days at 20 °C with different substrate and nutrient additions in a CarboBot system. For each site, depth (top- and subsoil) and sampling type (bulk and rhizosphere soil), three experimental replicates were set for control, soil + glucose (C₆H₁₂O₆) at a rate of 13.3 µg C mg⁻¹ SOC (Hamer and Marschner, 2005), soil + nitrogen (NH₄NO₃) with a rate of 110 mg N kg⁻¹ dry soil (Chen et al., 2014), soil + phosphorus (Ca(H₂PO₄)₂·H₂O - triple superphosphate) with a rate of 11 mg P kg⁻¹ dry soil and soil + palmitic acid (C₁₆H₃₂O₂) with a rate of 13.3 µg C mg⁻¹ SOC (Hamer and Marschner, 2005). N and P were used to elucidate the response of microorganisms to changing nutrient availability and identify potential nutrient limitations. Substrates were chosen based on their respective nominal oxidation state of carbon of 0 (glucose) and –1.75 (palmitic acid) (calculated according to LaRowe and van Cappellen (2011)) which quantifies the energy yield of the compound based on the Gibbs energies of the oxidation half reaction. A negative nominal oxidation state indicates a higher activation energy of the consumer with the prospect of higher energy yield (Spohn et al., 2016). We used ¹⁴C labeled glucose and palmitic acid (PerkinElmer, Waltham, USA) to differentiate between SOC and substrate C mineralization. Both ¹⁴C substrates were mixed with unlabeled substrates to reach a radioactivity of 20 kBq for each sample under the assumption that labeled and unlabeled substrate are mineralized equally. Since palmitic acid is only soluble in ethanol (EtOH), all other samples received each 2.36 µg C-EtOH µg⁻¹ substrate C to exclude any EtOH effects between the different treatments. For each sample, 40 g of dried sieved (<2 mm) soil was weight into 120 ml polyethylene flasks and was rewetted at 40% water holding capacity. Subsequently, the samples were put into airtight vessels with 10 ml of 0.5 M (topsoil) and 0.3 M (subsoil) potassium hydroxide (KOH) in a small beaker at the top of the vessels. All samples were pre-incubated for 5 days, allowing the microbial community to adapt to current conditions and minimize the rewetting effect. After pre-incubation, the samples received their specific compound solutions and were incubated for 82

days at 20 °C. The CO₂-evolution was measured continuously (every 4 h) through the detection of changes in electrical conductivity of the 0.5 M (topsoil) and 0.3 M (subsoil) KOH within a CarboBot (prw electronics, Berlin, Germany) unit. The KOH was renewed when a critical level of CO₂ saturation was reached. The emitted CO₂ from the ¹⁴C substrate additions was determined by using liquid scintillation counting. One ml of KOH was mixed with 6 ml scintillation cocktail (Ultima Gold, PerkinElmer, Waltham, USA) and measured with a Tri-Carb 2800 TR beta counter unit (PerkinElmer, Waltham, USA).

2.5. Microbial biomass

Microbial biomass C (C_{mic}) was determined with the chloroform fumigation extraction method (CFE) (Vance et al., 1987). Detailed information about the determination procedure is found in Herre et al. (2021). After 82 days of incubation, the CFE method was used again to determine ¹⁴C activity in the microbial biomass in the samples which received the glucose and palmitic acid treatments. From each K₂SO₄ filtrate, 2 ml were mixed with 12 ml scintillation cocktail (Ultima Gold, PerkinElmer, Waltham, USA) and the ¹⁴C-activity was measured by liquid scintillation counting. ¹⁴C in microbial biomass was calculated by subtracting the ¹⁴C-activity of non-fumigated samples from respective fumigated samples using a k_{EC} value of 0.45 (Wu et al., 1990). ¹⁴C_{mic} is presented as % of initial ¹⁴C-input. To account for the solvent fraction, K₂SO₄-extractable C and N were determined in the non-fumigated extracts.

2.6. Hydrolytic enzyme activity

To assess the response to changing substrate and nutrient supply, enzyme activities for different nutrient cycles were measured with a microplate fluorometric multi-substrate assay as described by Marx et al. (2001). β-glucosidase (BGL (C-Cycle)), N-Acetylglucosaminidase (NAC (C/N-Cycle)) and acid Phosphatase (PHO (P-Cycle)) were determined with a 4-methylumbelliferon (MUF) and L-Arginine-aminopeptidase (ARG (N-Cycle)) with a 7-amino-4-methylcoumarin (AMC) labeled substrate (Sigma-Aldrich Chemie GmbH, Munich, Germany). For each sample, 1 g of fresh soil were weighed into autoclaved beakers and filled up with 50 ml of ultrapure sterile water. The suspensions were sonicated for 1 min to release previously immobilized enzymes from soil particles (Marx et al., 2001). To reach the desired pH optimum for the enzymes, a 0.1 M MES buffer (pH 6.1) for the MUF substrates and 0.05 M TRIZMA buffer (pH 7.8) for the AMC substrate was added. The micro plates were pre-incubated for 10 min at 30 °C to reach the temperature optimum for each enzyme. The fluorescence was measured at 0, 30, 60, 120 and 180 min with a Tecan infinite 200 microplate reader (Tecan Trading AG, Mannedorf, Switzerland) unit at 360 nm excitation and 465 nm emission wavelength. Enzyme activity was calculated as the development of the fluorescence over time (180 min) in nM g⁻¹ h⁻¹. To determine changes in enzyme production through substrate and nutrient addition, enzyme activity was measured prior and after 82 days of incubation.

2.7. Oxidoreductase enzyme activities

Dehydrogenase activity (DEH) was determined as a proxy for active microbial biomass as this intracellular enzyme is only produced by living organisms (Wolinska and Stepniowski, 2012). The method was

based on Mersi and Schinner (1991) and modified using iodonitrotriazolium chloride (INT) as substrate which is reduced to iodonitrotriazolium formazan (INTF) by the dehydrogenase enzyme. For each sample, 1 g of fresh soil was weighed into a 15 ml centrifuge tube and mixed with 0.5 ml 1 M TRIS buffer (pH 7) and 0.5 ml aqueous INT-solution with a rate of 4 mg ml⁻¹. Blanks received each 0.5 ml H₂O instead of INT. Subsequently, the suspensions were shaken with a Vortex mixer and incubated for 2 h at 40 °C. After 2 h of incubation, the enzymatic reaction was interrupted by adding 5 ml methanol. The samples were then shaken (Vortex mixer) and maintained for an additional 60 min at 20 °C. Finally, the samples were centrifuged at 3000 rpm for 10 min, and the supernatants were pipetted into 96-well plates. The absorption was measured at 492 nm wavelength using a TECAN infinite 200 fluorescence multifunctional microplate reader (TECAN Group, Mannedorf, Switzerland). DEH was calculated as μg INTF g⁻¹ SOC h⁻¹.

Peroxidase activity (POX) was measured as this oxidative enzyme is produced by organisms in order to depolymerize recalcitrant compounds such as lignin and can be used as proxy for the capability of microorganisms to utilize more complex substrates (Sinsabaugh, 2010; Burns et al., 2013). The method was used in modification to Sinsabaugh et al. (2003) using L-3,4-dihydroxyphenylalanine (DOPA) as substrate. For each sample, 0.4 g fresh soil was weighed into 100 ml beakers and received each 50 ml of 50 mM acetate buffer. The soil suspensions were continuously stirred using a magnetic stir plate. Subsequently, 200 μl of the stirring soil suspension was dispensed in 96-well plates and received 50 μl 25 mM DOPA and 10 μl H₂O₂ (0.3%). Negative control and blank wells were also prepared as described by Sinsabaugh et al. (2003). The samples were incubated for 24 h at 20 °C in the dark. After incubation, the peroxidase activity was quantified by measuring absorbance at 450 nm using a TECAN infinite 200 fluorescence multiplate reader (TECAN Group, Mannedorf, Switzerland). POX was calculated as μM h⁻¹ SOC.

2.8. Statistical analysis

Statistical analysis was performed with R-Studio (Version 1.0.143) with the in-built and additional libraries Stats (Version 3.6.0), Rmisc (V. 1.5) and agricolae (V. 1.2.8). For each site, the data was grouped into "Bulk" and "Rhizosphere" soil subsets and divided into topsoil ("0–10") and subsoil ("30–50"), respectively. All subsets were tested for normal distribution using the Shapiro-Wilk test. Subsets with a significant test statistic W (P-value > 0.05) were considered normal distributed. All other subsets were log-transform to achieve normality and homoscedasticity. To identify potential priming effects, significant differences between the evolved unlabeled C–CO₂ from the glucose and palmitic acid treatments and the evolved C–CO₂ from the control sample were determined. Significant differences were determined using a one-way ANOVA with a post-hoc Tukey-HSD-test (P < 0.05). The SOC mineralization (SOC_{min}) of the ¹⁴C-glucose and ¹⁴C-palmitic acid amended samples was calculated as:

$$SOC_{min} = Total\ CO_2 - C_{treatment} - {}^{14}CO_2 - C$$

where the emitted CO₂–C from the ¹⁴C amended samples was subtracted from the total CO₂–C of the sample. The priming intensity (PI_t) in % of the initial substrate input was calculated for each time interval (Δt) according to Hamer and Marschner (2002) as:

$$PI_t [\% \text{ of initial input } d^{-1}] = 100 \cdot \frac{(\Delta Total\ CO_2 - C_{treatment} - \Delta {}^{14}CO_2 - C) - \Delta CO_2 - C_{control}}{\Delta CO_2 - C_{control} \cdot \Delta t}$$

Table 1
Basic soil properties: Mean values of three field replicates for bulk and rhizosphere topsoil and subsoil at all sampling sites. Standard deviation ($n = 3$) is presented in brackets. Significant differences between bulk and rhizosphere soil within the same depth are shown by the following symbols: “*” = $p < 0.05$, “**” = $p < 0.01$ and “***” = $p < 0.001$. Data from Herre et al. (2021).

Site	Depth	Clay (%)		Silt (%)		Sand (%)		pH		SOC (%)		N _t (%)		C _{mic} (mg kg ⁻¹)	
		Bulk		Rhizosphere		Bulk		Rhizosphere		Bulk		Rhizosphere		Bulk	
PS	0–10 cm	3.87 (1.15)	42.69 (8.69)	53.44 (9.23)	2.98 (0.09)	2.93 (0.03)	1.75 (0.46)	7.39 (0.80) ***	0.12 (0.07)	0.23 (0.05)	213.2 (146.28)	409.75 (114.04)			
	30–50 cm	4.84 (0.71)	43.52 (11.72)	51.64 (12.39)	4.11 (0.15)	4.05 (0.15)	0.55 (0.07)	1.49 (0.23) **	0.03 (0.00)	0.07 (0.01) **	17.57 (4.64)	127.76 (25.28) **			
LS	0–10 cm	10.32 (1.89)	88.94 (1.33)	0.74 (0.56)	4.02 (0.10)	4.38 (0.26)	2.34 (0.37)	6.77 (2.16) *	0.18 (0.02)	0.49 (0.16) *	289.63 (55.07)	1026.73 (84.03) *			
	30–50 cm	9.12 (0.29)	90.37 (0.18)	0.51 (0.15)	3.87 (0.11)	3.81 (0.02)	0.22 (0.04)	0.70 (0.13) **	0.03 (0.00)	0.06 (0.01) **	6.22 (0.00)	73.21 (30.23) **			
TS	0–10 cm	6.76 (0.34)	48.81 (1.64)	44.42 (1.92)	3.94 (0.40)	4.35 (0.03)	3.71 (1.23)	4.47 (0.32)	0.23 (0.07)	0.28 (0.03)	192.22 (71.66)	777.57 (98.73) **			
	30–50 cm	10.87 (0.64)	51.34 (10.07)	37.79 (9.99)	3.92 (0.14)	4.11 (0.06)	0.21 (0.09)	1.21 (0.25) **	0.03 (0.01)	0.08 (0.01) *	13.07 (5.70)	158.99 (37.43) **			

where $\Delta\text{Total CO}_2 - \text{C}_{\text{treatment}}$ is the cumulative amount of mineralized C in the glucose and palmitic acid amended samples, $\Delta^{14}\text{CO}_2 - \text{C}$ is the substrate-derived $\text{CO}_2 - \text{C}$ and $\Delta\text{CO}_2 - \text{C}_{\text{control}}$ is the cumulative amount of mineralized SOC in the respective control samples during each time interval t . A multivariate analysis of variance (MANOVA) was conducted for testing significant influences of sampling site, sampling type and sampling depth for selected parameters. A principal component analysis (PCA) was carried out to analyze the relations between parameters within in the same sampling depth.

3. Results

3.1. Soil properties, SOC, microbial biomass

At all sample sites, bulk density increased along the profile and ranged between 1.02 and 1.2 g cm⁻³ in the topsoil and between 1.34 and 1.49 g cm⁻³ in the subsoil, respectively. Soil texture of the PS soil was characterized by low clay and dominated by the sand fraction in both topsoil and subsoil. The LS site was dominated by the silt fraction with 88.9% in the topsoil and 90.4% in the subsoil, respectively and clay content of 9.12–10.32%. The TS site mainly consists of silt sized material with increasing values with increasing depth and low clay content in the topsoil that increased in the subsoil.

SOC content was higher in the rhizosphere soil compared to the respective bulk samples at all sample sites and in both topsoil and subsoil (Table 1). Total N decreased from topsoil to subsoil at all sample sites with significant differences in the subsoil with higher values in the rhizosphere soil compared to the respective bulk samples. Similar to the SOC content, C_{mic} was higher in the rhizosphere soil compared to the respective bulk soil throughout all soil profiles with significant differences in all subsoil samples. In average, C_{mic} content was 3.1-fold and 10.4-fold higher in the rhizosphere topsoil and subsoil compared to the respective bulk samples. K_2SO_4 -extractable C as a proxy for the soluble fraction was higher in all rhizosphere samples compared to the respective bulk soil with significant differences in the topsoil at the PS and LS site (Table 3). When normalized to SOC, higher K_2SO_4 -extractable C in all subsoil samples compared to the respective topsoil indicates a higher proportion of labile C.

3.2. SOC mineralization and microbial C use efficiency

During the 82 days of incubation, the total SOC mineralization in the control samples ranged from 35 to 91 mg $\text{CO}_2 - \text{C g}^{-1}$ SOC in the topsoil and 15–181 mg $\text{CO}_2 - \text{C g}^{-1}$ SOC in the subsoil, reflecting a high variability but also a large proportion of labile components in all samples (Fig. 1). At the PS site, the SOC mineralization in the bulk soils was 14% lower in the subsoil than in the topsoil but in the rhizosphere soils, 35% higher in the subsoil than in the topsoil. At the LS site, SOC mineralization in the subsoil bulk samples showed the highest values of all samples and was 2-fold higher than in the topsoil, while the SOC mineralization in the rhizosphere was only 14% higher in the subsoil compared to the topsoil rhizosphere. At the TS site, the SOC mineralization showed a different pattern with 31% and 36% less mineralization in both the subsoil bulk and rhizosphere soil compared to the respective topsoil (Fig. 1). After N addition, we observed a general trend of decreasing SOC mineralization compared to the control sample with a significant reduction in the TS bulk topsoil and at the PS site in both depths and sampling types (i.e., bulk and rhizosphere soil). The addition of N + P caused a similar reaction and also showed a trend of declining mineralization compared to the control with significant effects in the PS topsoil, TS bulk topsoil and PS rhizosphere subsoil. Phosphorus addition also decreased the SOC mineralization with significant differences to the control at the PS site in both bulk and rhizosphere soil.

The addition of glucose increased the mineralization in all samples from the LS and TS sites, thus inducing positive priming effects. In contrast, a decrease in SOC mineralization, i.e., negative priming was

Table 2

Relative enzyme activity (normalized to SOC content) from the different treatments at the end of the incubation for β -Glucosidase (BGL), N-Acetylglucosaminidase (NAC), Acid Phosphatase (PHO) and L-Arginine-aminopeptidase (ARG) in bulk and rhizosphere soil for topsoil and subsoil. Standard deviation (n = 3) is presented in brackets (n. d. = not detectable).

Site	Depth	Treatment	L-Arginine-aminopeptidase ($\mu\text{mol g}^{-1} \text{SOC h}^{-1}$)		β -Glucosidase ($\mu\text{mol g}^{-1} \text{SOC h}^{-1}$)		N-Acetylglucosaminidase ($\mu\text{mol g}^{-1} \text{SOC h}^{-1}$)		Acid Phosphatase ($\mu\text{mol g}^{-1} \text{SOC h}^{-1}$)	
			Bulk	Rhizosphere	Bulk	Rhizosphere	Bulk	Rhizosphere	Bulk	Rhizosphere
PS	0–10 cm	C	8.91 (6.87)	3.52 (1.58)	12.82 (2.4)	57.79 (22.2)	17.43 (3.5)	85.16 (14.52)	421.73 (97.73)	404.28 (79.89)
		GL	10.19 (4.1)	3.83 (1.04)	n. d.	20.26 (1.55)	13.35 (3.06)	28.05 (18.51)	280.84 (48.45)	352.19 (32.92)
		N	4.13 (1.76)	n. d.	n. d.	13.32 (2.38)	n. d.	20.36 (11.8)	113.46 (28.26)	277.27 (21.56)
		NP	n. d.	n. d.	n. d.	12.08 (1.53)	n. d.	18.42 (3.75)	102.78 (31.91)	272.54 (6.78)
		P	8.42 (3.59)	1.96 (0.24)	n. d.	21.93 (4.68)	7.38 (2.85)	30.48 (14.89)	182.03 (18.27)	310.53 (26.68)
		PA	7.9 (3.02)	2.26 (0.36)	42.67 (26.45)	60.43 (9.14)	44.14 (11.52)	79.68 (16.07)	193.92 (20.26)	325.22 (38.81)
PS	30–50 cm	C	n. d.	9.71 (1.58)	17.15 (6.39)	58.93 (26.26)	17.17 (5.65)	62.08 (19.66)	340.84 (85.62)	483.23 (100.2)
		GL	13.1 (7.64)	8.59 (1.43)	19.11 (11.41)	37.67 (8.93)	26.21 (11.94)	97.99 (43.95)	374.77 (94.87)	506.21 (137.51)
		N	n. d.	n. d.	12.88 (2.4)	23.01 (13.64)	n. d.	45.24 (18.04)	111.84 (42.27)	312.37 (65.82)
		NP	5.5 (0.84)	n. d.	18.65 (3.56)	24.04 (11.32)	n. d.	41.37 (20.08)	116.72 (41.32)	340.59 (20.28)
		P	9.9 (4.49)	5.14 (1.07)	17.06 (3.68)	41.8 (10.75)	19.89 (5.96)	78.49 (43.35)	219.88 (31.01)	362.38 (99.74)
		PA	n. d.	n. d.	31.76 (2.83)	66.22 (6.52)	40.93 (11.79)	109.79 (38.54)	448.73 (157.52)	571.65 (83.02)
LS	0–10 cm	C	30.08 (4.35)	19.24 (4.12)	116.3 (0.66)	104.16 (32.75)	44.28 (9.27)	73.59 (29.37)	1164.9 (281.71)	721.33 (163.06)
		GL	35.91 (4.18)	16.15 (1.37)	102.52 (10.22)	112.78 (12.72)	60.78 (7.5)	94.42 (46.31)	1154.8 (246.57)	784.51 (165.05)
		N	8.14 (0.77)	20.72 (5.49)	76.67 (38.57)	102.42 (9.9)	43.52 (22.04)	77.44 (31.12)	1667.07 (670.35)	818.44 (139.65)
		NP	n. d.	20.08 (6.6)	74.41 (20.09)	133.66 (69.25)	51.97 (29.75)	100.91 (63.62)	1191.86 (305.2)	787.15 (185.82)
		P	26.64 (8.99)	25.47 (6.59)	130.21 (22.91)	102.89 (25.95)	56.27 (4.5)	89.84 (33.85)	969.92 (161.05)	631.15 (148.7)
		PA	45.5 (4.3)	21.31 (2.54)	116.22 (36.97)	136.91 (38.95)	65.13 (22.38)	96.97 (33.34)	1259.77 (446.97)	788.49 (168.02)
LS	30–50 cm	C	n. d.	64.3 (16.57)	31.3 (11.61)	205.06 (196.2)	29.93 (4.62)	120.3 (68.14)	692.91 (158.85)	906.8 (671.87)
		GL	n. d.	46.96 (10.3)	33.86 (10.69)	106.13 (21.9)	n. d.	101.51 (58.86)	726.7 (282.26)	918.41 (471.51)
		N	n. d.	n. d.	112.04 (65.81)	66.6 (23.73)	53.35 (38.52)	69.75 (52.56)	6128.99 (1793.9)	2105.57 (769.24)
		NP	n. d.	n. d.	240.86 (264.69)	168.28 (132.06)	152.68 (173.83)	145.06 (66.89)	1151.07 (848.16)	725.71 (225.85)
		P	n. d.	42.66 (24.23)	28.51 (10.54)	131.07 (22.77)	25.1 (4.19)	101.02 (30.22)	596.45 (188.6)	949.68 (435.55)
		PA	n. d.	73.15 (24.2)	26.41 (9.36)	144.07 (136.34)	n. d.	113.3 (66.85)	673.54 (56.65)	828.56 (314.04)
TS	0–10 cm	C	20.86 (10.02)	11.48 (1.3)	79.51 (30.69)	117.02 (98.54)	55.67 (9.66)	117.13 (52.06)	698.56 (170.73)	975.04 (194.39)
		GL	28.83 (17.93)	12.44 (2.64)	51.55 (4.51)	50.69 (11.17)	77.43 (26.21)	75.26 (12.55)	493.23 (13.99)	674.85 (94.76)
		N	10.86 (9.09)	26.32 (9.61)	64.07 (21.96)	109.08 (56.33)	50.77 (16.09)	106.47 (11.38)	794.61 (274.83)	1108.91 (41.47)
		NP	13.21 (3.24)	22.18 (4.89)	49.28 (8.8)	113.58 (61.16)	48.82 (6.3)	141.25 (24.3)	724.32 (233.3)	983.76 (103.67)
		P	24.34 (4.82)	14.46 (0.94)	70.05 (24.16)	67.87 (12.01)	73.62 (31.48)	85.91 (11.57)	772.67 (233.27)	918.55 (59.31)
		PA	23.91 (7.72)	17.05 (6.04)	54.4 (14.79)	69.35 (9.25)	54.02 (2.79)	80.09 (18.82)	696.45 (139.97)	781.01 (84.83)
TS	30–50 cm	C	n. d.	47.09 (16.96)	44.27 (21.1)	117.57 (43.61)	123.36 (52.69)	111.89 (48.94)	695.33 (267.78)	373.13 (47.34)
		GL	n. d.	50.72 (8.74)	46.09 (35.33)	87.24 (28.8)	49.04 (30.23)	84.85 (20.72)	971.27 (542.07)	502.4 (78.89)
		N	n. d.	n. d.	n. d.	45.23 (28.32)	n. d.	51.04 (11.36)	290.08 (130.63)	294.6 (49.29)
		NP	n. d.	n. d.	24.68 (9.32)	45.2 (32.16)	n. d.	56.52 (16.25)	314.47 (106.22)	256.31 (32.12)
		P	n. d.	33.07 (12.25)	37.17 (9.75)	108.15 (62.08)	190.52 (171.9)	121.61 (25.87)	618.35 (264.64)	466.5 (31.24)
		PA	n. d.	57.76 (8.93)	74.05 (58.48)	101.59 (49.11)	241.45 (167.57)	100.48 (33.15)	1009.16 (377.58)	762.55 (50.98)

observed at the PS site in the bulk topsoil and rhizosphere subsoil. Palmitic acid also stimulated SOC mineralization at the LS and TS sites. Interestingly, the SOC mineralization rates after palmitic acid addition were similar or even higher than that after glucose addition.

The efficiency of microbial C use and C availability was determined using the metabolic quotient ($q\text{CO}_2$) ($\text{mg CO}_2\text{-C g C}_{\text{mic}}^{-1} \text{h}^{-1}$) and microbial biomass C to SOC ratio ($\text{C}_{\text{mic}}:\text{SOC}$). With the exception of the PS topsoil, the $q\text{CO}_2$ was lower in all rhizosphere samples compared to the

Table 3

Metabolic quotient (qCO_2), C_{mic} :SOC ratio, Peroxidase activity (POX) and K_2SO_4 extractable C. Mean values of three field replicates for bulk and rhizosphere topsoil and subsoil at all sampling sites. Standard deviation ($n = 3$) is presented in brackets. Significant differences between bulk and rhizosphere soil within the same depth are shown by the following symbols: *** = $p < 0.05$, **** = $p < 0.01$ and ***** = $p < 0.001$. Data for C_{mic} :SOC and K_2SO_4 -extractable C taken from [Herre et al. \(2021\)](#).

Site	Depth	qCO_2 (mg CO_2 -C g C_{mic}^{-1} h $^{-1}$)		C_{mic} :SOC (%)		K_2SO_4 - extractable C (mg kg $^{-1}$)		K_2SO_4 - extractable C (mg kg $^{-1}$ SOC)		POX (μ M g SOC $^{-1}$ h $^{-1}$)	
		Bulk	Rhizosphere	Bulk	Rhizosphere	Bulk	Rhizosphere	Bulk	Rhizosphere	Bulk	Rhizosphere
PS	0–10 cm	2.65 (1.26)	5.29 (1.56)	1.23 (0.85)	0.56 (0.16)	47.05 (10.89)	70.69 (2.46) *	27.40 (7.43)	9.61 (0.69) **	200.81 (31.99)	30.82 (6.51) ***
	30–50 cm	7.92 (1.35)	2.93 (0.90) **	0.33 (0.10)	0.87 (0.23) *	49.96 (11.62)	74.06 (10.88)	92.37 (27.91)	51.56 (18.21) *	255.29 (148.75)	104.75 (61.05)
LS	0–10 cm	2.41 (0.45)	2.07 (0.22)	1.25 (0.26)	1.50 (0.22)	32.93 (4.98)	59.25 (9.1) *	14.03 (0.71)	9.06 (1.60) **	147.43 (23.11)	39.99 (27.43) **
	30–50 cm	24.31 (3.64)	3.13 (1.42) ***	0.28 (0.04)	1.02 (0.25) **	32.41 (8.02)	44.52 (2.38)	143.43 (20.21)	65.75 (14.65)	2229.84 (430.88)	539.78 (158.20) **
TS	0–10 cm	4.58 (3.44)	1.16 (0.12)	0.61 (0.44)	1.74 (0.10) *	35.82 (11.52)	70.49 (24.31)	11.07 (6.58)	15.53 (5.42)	74.16 (45.32)	57.90 (9.20)
	30–50 cm	2.09 (1.20)	0.98 (0.05)	0.69 (0.43)	1.31 (0.12)	27.44 (16.18)	59.38 (13.47)	124.01 (46.83)	51.81 (21.48)	2494.51 (612.93)	286.24 (42.82) **

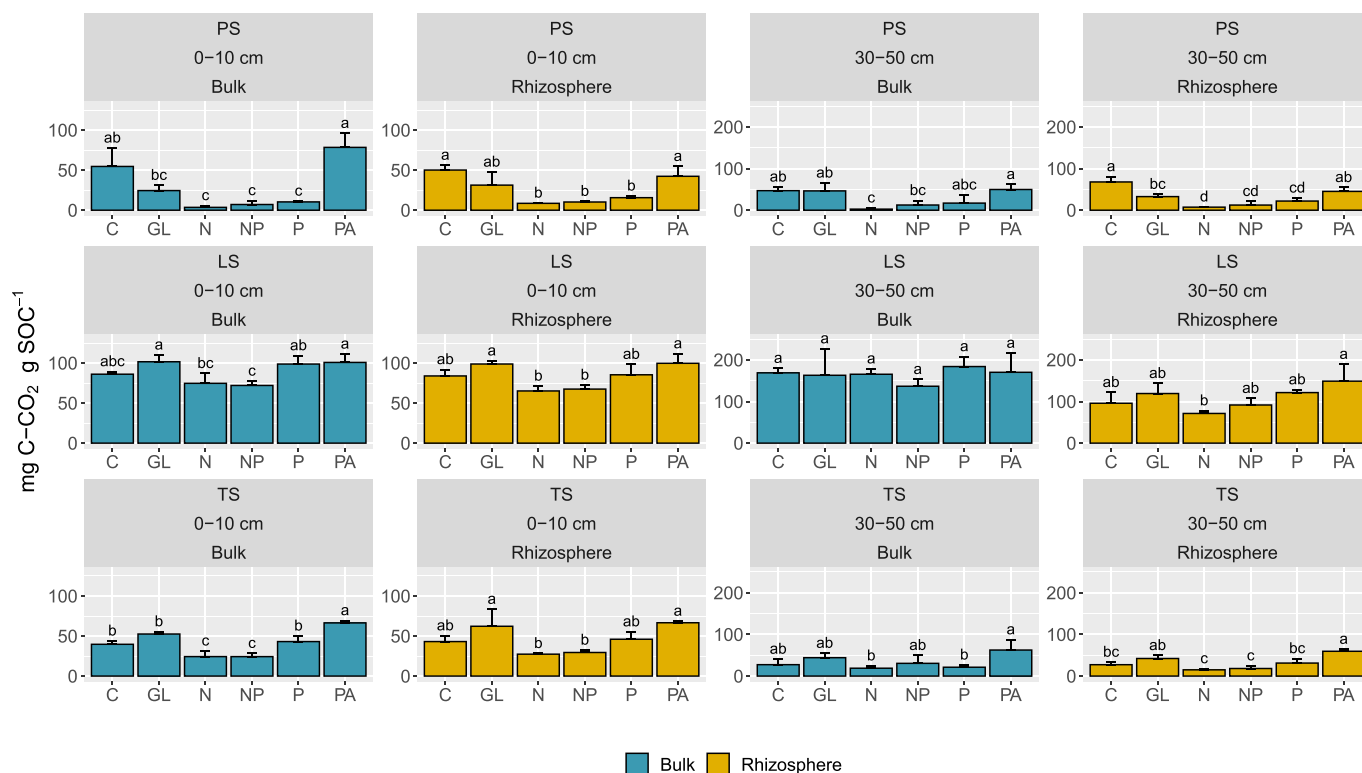


Fig. 1. Cumulative SOC derived CO_2 -C (mg C- CO_2 g SOC $^{-1}$) after 82 days of incubation in bulk and rhizosphere soil from the top and subsoil. Please note that topsoil and subsoil have different scales. C = Control, GL = glucose, N = nitrogen, NP = nitrogen + phosphorus, P = phosphorus and PA = palmitic acid. Error bars indicate standard deviation of replicates ($n = 3$). Different letters within the same depth indicate significant differences between treatments ($P < 0.05$).

respective bulk soil with significant differences in the PS and LS subsoil indicating a higher carbon use efficiency in soil closely associated with the roots (Table 3). Likewise, the C_{mic} :SOC ratio was always higher in the rhizosphere soil compared to the bulk soil with significant differences in the TS topsoil and PS and LS subsoil (Table 3).

3.3. Priming intensity

The priming intensity (PI_t) in the topsoil for glucose additions over time showed the highest effects within the first few days of incubation at all sampled sites and depths. Interestingly, the PI_t at the PS site fell below 0% right after the start of the incubation and stayed negative in

the bulk topsoil throughout the entire incubation procedure. In the rhizosphere soil however, PI_t rose quickly afterwards and peaked with 7.3% around day 10 of the incubation. This pattern was also observed at the LS site with a peak around day 10 and a steady decline in PI_t towards the end of the incubation. At the TS site, PI_t peaked right in the beginning of the incubation with 61% (bulk) and 96% (rhizosphere) after glucose addition and steadily declined towards the end of the incubation with final values of $<1\%$ (Fig. 2). In the PS subsoil, the respective PI_t reached the lowest values right at the beginning of the incubation and continued to be negative in the rhizosphere soil. At the LS site, we observed a contrariwise curve between the two sampling types with a negative peak at day 7 in the bulk soil (-4.97%) and a positive peak at

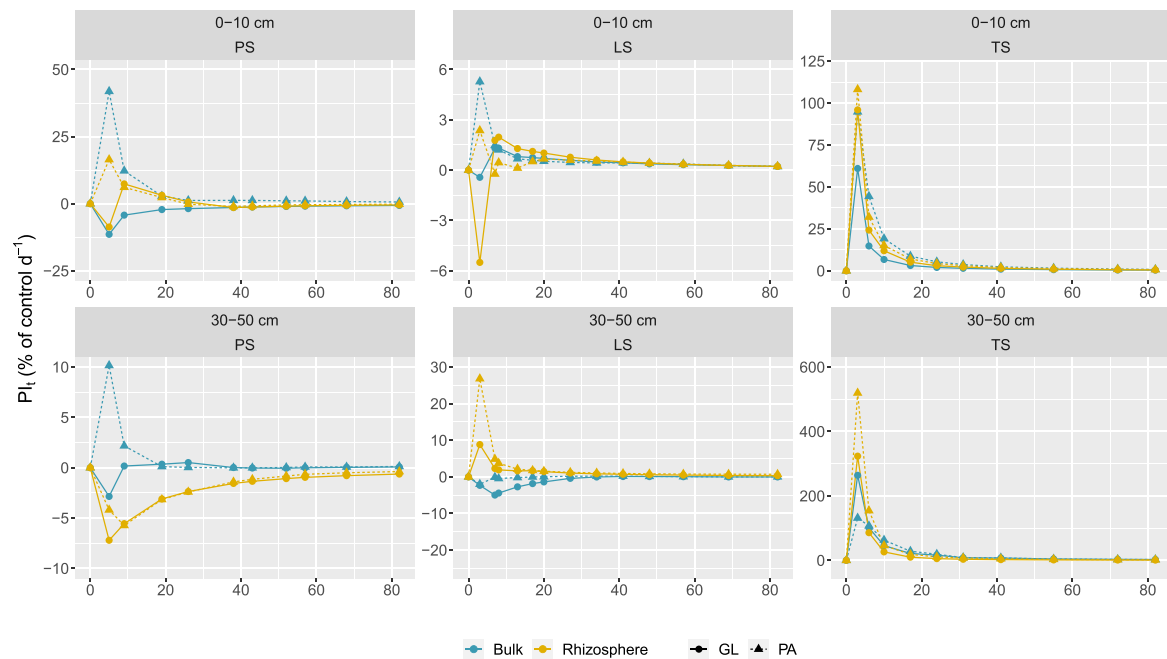


Fig. 2. Priming intensity [% of control d⁻¹] during 82 days of incubation in bulk (BU) and rhizosphere (RH) soil from the top and subsoil. GL = ¹⁴C labeled glucose and PA = ¹⁴C labeled palmitic acid.

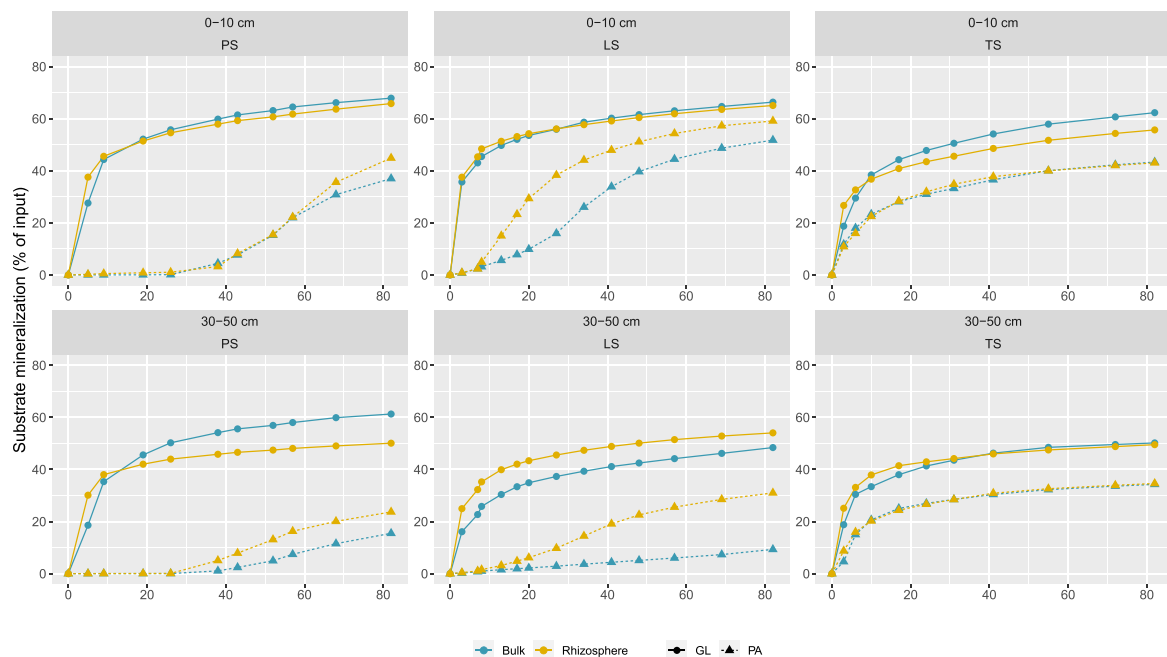


Fig. 3. Substrate mineralization in % of initial input during 82 days of incubation in bulk (BU) and rhizosphere (RH) soil from the top and subsoil. GL = ¹⁴C labeled glucose and PA = ¹⁴C labeled palmitic acid.

day 3 in the rhizosphere soil (8.83%). Interestingly, the PI_t in the TS subsoil showed a similar curve progression compared to the topsoil with a peak at day 3 (264% bulk, 323% rhizosphere) and a steady decline afterwards. In contrast to glucose, palmitic acid addition in the topsoil led to positive values at all sites with significant peaks within the first days of the incubation. Interestingly, PI_t was higher in the bulk soil compared to the respective rhizosphere samples at the PS (42% bulk, 16% rhizosphere) and LS (5.3% bulk, 2.4% rhizosphere) site. At the TS site, the addition of palmitic acid showed a similar pattern as the glucose

addition with a pronounced peak of 95% (bulk) and 108% (rhizosphere) in the topsoil after five days and a steady decline afterwards (Fig. 2). In the subsoil, palmitic acid addition resulted in a positive peak at the beginning of the incubation in the PS bulk soil (10.1%) with a steady decline over time and a contrary progression in the rhizosphere soil with the lowest values (−4.2%) at day 5 and a steady increase towards the end of the incubation. At the LS site, PI_t showed a pronounced peak in rhizosphere soil (26.8%) within the first days, while palmitic acid addition in the bulk soil had no significant effect. Similar to the topsoil,

PI_t peaked at the TS site in the rhizosphere (520%) and bulk soil (131%) at the beginning of the incubation and declined afterwards over the course of 82 days of incubation (Fig. 2).

3.4. Substrate mineralization

The substrate mineralization over time showed a rapid mineralization of the added ¹⁴C-labeled glucose in both topsoil and subsoil at all sites and depths (Fig. 3). Within the first 10 days of incubation, already 36.8–48.4% of the added ¹⁴C-glucose were mineralized in the bulk and rhizosphere topsoil samples. In the subsoil, glucose mineralization rates were generally slightly lower and reached 25.8–38.0% within the first 10 days. Interestingly, the ¹⁴C-glucose mineralization at the TS site showed higher values in the rhizosphere subsoil compared to the respective topsoil samples. The mineralization of ¹⁴C-labeled palmitic acid showed no clear pattern among the three sampling sites. After 82 days, the cumulative mineralization rate of palmitic acid in the topsoil reached 45% in the rhizosphere compared to only 37% in the bulk soil. Similarly in the subsoil, mineralization rates were significantly higher in the rhizosphere (23.7%) compared to 15.5% in the bulk soil. At the LS site, palmitic acid mineralization was more rapid, surpassing 5% and 3% within the first 10 days of incubation and reaching a total of 59.1% and 51.7% after 82 days ($P < 0.05$) in the topsoil rhizosphere and bulk soil, respectively. In the LS subsoil, the mineralization of palmitic acid was delayed but ascended rapidly after day 10 in the rhizosphere and reached a total of 31% and 9.3% in the rhizosphere and bulk soil, respectively ($P < 0.05$). At the TS site, the mineralization of palmitic acid showed no lag period in any sample and the cumulative rates reached about 43% in the topsoil samples and 34% in the subsoil samples, independent of sampling type.

3.5. Hydrolytic enzyme activities

Similar to the SOC mineralization rates, the measured activities of enzymes were normalized to SOC to account for the high variation of SOC content within the same depth between the three sites and sampling types (i.e., bulk and rhizosphere soil) and termed “relative enzyme activities”. At the PS site, enzyme activity were distinctly affected by substrate and nutrient additions as well as sampling depth. The addition of N and the combination of N + P led to a decrease in enzyme activity throughout all treatments and depths with some samples below detection limit (Table 2). P addition showed no constant response in enzyme activity. The responses of enzyme activity to the substrate additions of glucose and palmitic acid were strongly affected by sampling depth. In the topsoil, glucose slightly increased ARG in bulk and rhizosphere soil while it was below detection limit in most subsoil samples. BGL and NAC showed a similar pattern in the topsoil with a decrease after glucose addition in bulk and rhizosphere and a significant increase after palmitic acid addition in the topsoil (Fig. 4). PHO activity was negatively influenced by glucose and palmitic acid addition in the topsoil. At the LS site, N addition led to a decrease in ARG and BGL activity in the bulk topsoil while no differences in activity were observed for the topsoil rhizosphere for all enzymes. In the subsoil, ARG was below detection limit in the bulk soil in all treatments. PHO activity in the topsoil was not significantly affected by substrate or nutrient addition, although N addition led to an increase in activity in the bulk soil. This increase was also observed in the subsoil, where PHO activity in the bulk soil was 8.7-fold higher compared to the control sample.

3.6. Oxidoreductase enzyme activities

After 82 days of incubation, the SOC normalized dehydrogenase activity was mostly affected by nutrient additions. In the PS topsoil, N and N + P treatments caused a significant decrease in activity in both

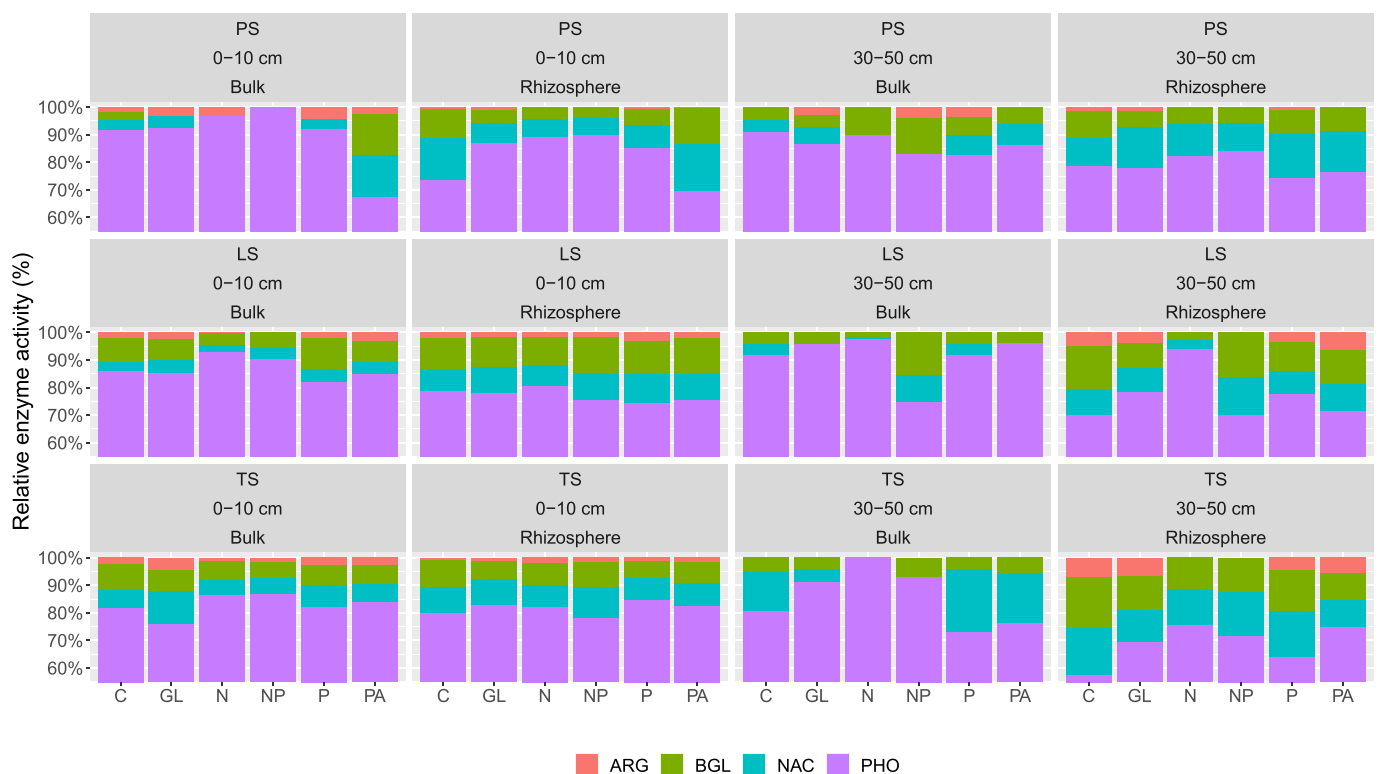


Fig. 4. Relative contribution of β -Glucosidase (BGL), N-Acetylglucosaminidase (NAC), Acid Phosphatase (PHO) and L-Arginine-aminopeptidase (ARG) to total enzyme activity in bulk and rhizosphere soil for topsoil and subsoil.

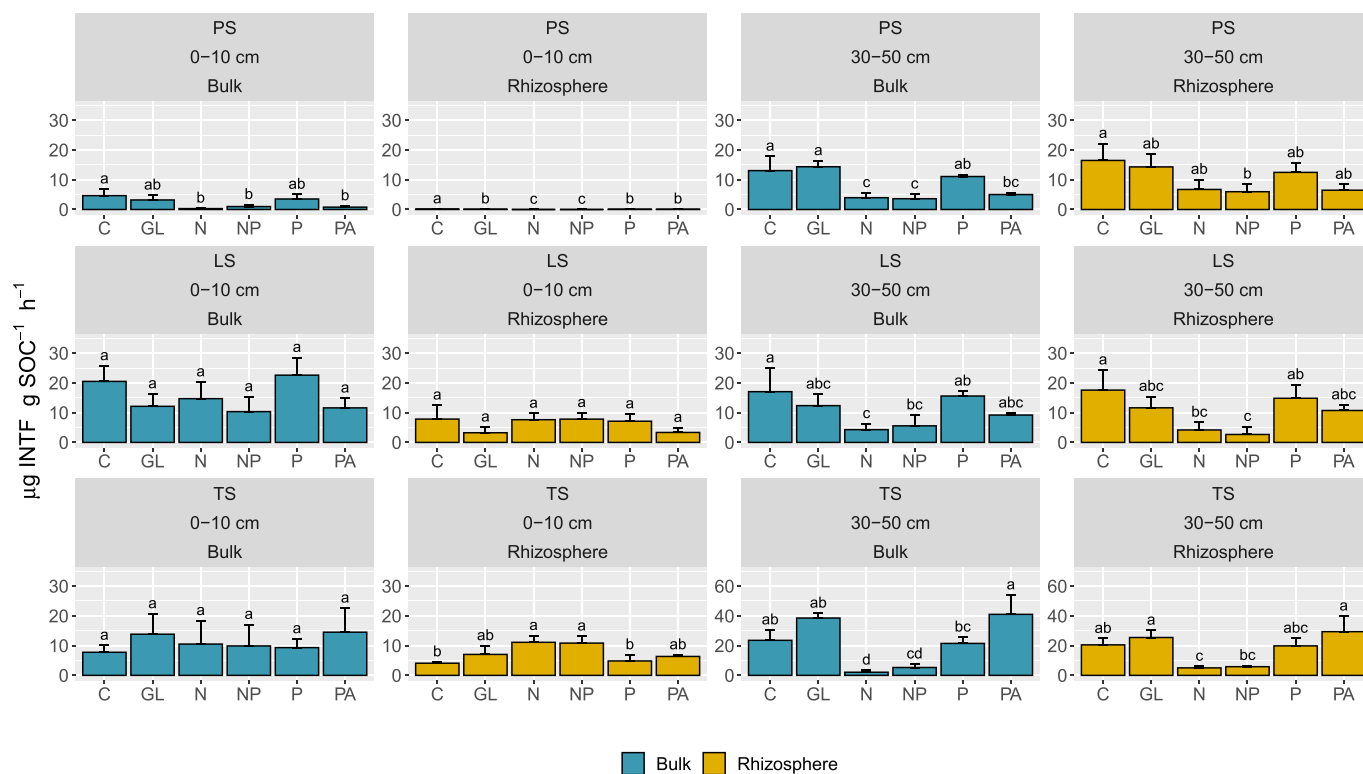


Fig. 5. Dehydrogenase activity ($\mu\text{g INTF g SOC}^{-1} \text{h}^{-1}$) after 82 days of incubation in bulk and rhizosphere soil from the top and subsoil. C = Control, GL = ^{14}C labeled glucose, N = nitrogen, NP = nitrogen + phosphorus, P = phosphorus and PA = palmitic acid. Error bars indicate standard deviation of replicates ($n = 3$). Different letters within the same depth indicate significant differences ($P < 0.05$).

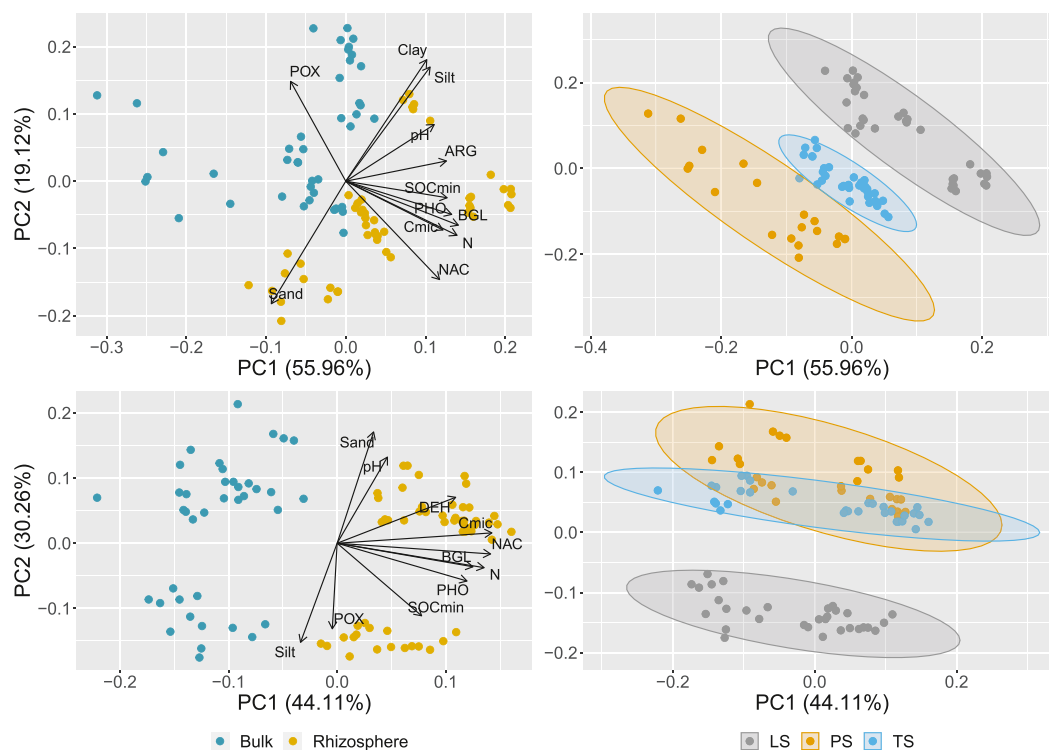


Fig. 6. Results of the principal component analysis PC1 and PC2 for topsoil (top) and subsoil (bottom) samples. Clustering of the dataset was based on sampling type (left) and sampling site (right).

bulk and rhizosphere soil (Fig. 5). While no effects of nutrient or substrate additions were observed in the LS topsoil, the TS site once again showed different patterns, with a significant increase of dehydrogenase activity in the rhizosphere topsoil in the N and N + P treatments. In the subsoil, N and N + P additions led to a significant decrease in both bulk and rhizosphere soil at all sample sites.

The addition of glucose led to an inconsistent pattern in DEH. At the PS site, DEH was significantly reduced in the rhizosphere topsoil. Palmitic acid addition induced a significant ($P < 0.05$) decrease of DEH in the PS topsoil and bulk subsoil and notably less activity in the rhizosphere subsoil. In correspondence to glucose addition, a general trend of DEH reduction at the LS site and increase at the TS site was observed.

SOC normalized peroxidase activity was not affected by nutrient or substrate additions at any site or depth. Interestingly, a higher POX activity was generally observed in the bulk samples compared to the respective rhizosphere samples in both topsoil and subsoil (Table 3). Significant differences between bulk and rhizosphere soil were found in the PS ($P < 0.001$) and LS ($P < 0.01$) topsoil. In general, peroxidase activity was higher in the subsoil compared to the topsoil with 1-fold (PS), 15-fold (LS) and 26-fold (TS) more $\mu\text{M g SOC}^{-1} \text{ h}^{-1}$ in the respective bulk samples. Significant differences between bulk and rhizosphere soil in the subsoil were found at the LS site ($P < 0.05$) (Table 3).

3.7. Depth specific relationships between parameters

A PCA was carried out to analyze the relations between parameters within in the same sampling depth. For each depth, two principal components were extracted only including parameters with a Kaiser-Meyer-Olkin-Criteria (KMO) > 0.5 . For the topsoil, PC1 explained about 56% of the variance, and with the highest factor loadings for N content, PHO and BGL. PC2 (19%) was loaded with POX, Clay and Sand (Supplementary Table 1). A clustering of the dataset regarding the investigated sample sites is visible with a clear differentiation of the PS site and coinciding data points of the LS and TS site (Fig. 6). In the subsoil, PC1 explained 44% and was positive coupled with C_{mic} , N content and enzymes. PC2 is characterized by texture, pH and POX. Different to the topsoil, there is a clear clustering of the dataset into the sampling types (i.e., bulk and rhizosphere soil) and a clear separation of the LS site (Fig. 6).

4. Discussion

4.1. SOC mineralization

Total SOC mineralization in the control samples was 29–296% (topsoil) and 73–563% (subsoil) higher in the rhizosphere samples compared to the respective bulk soil. These results are in line with Cheng et al. (2003) who found 60–383% higher SOC mineralization in the rhizosphere soil compared to root free soil and a recent meta-analysis which calculated an average 59% higher SOC mineralization in rhizosphere soil compared to bulk soil (Huo et al., 2017). In our case, the SOC normalized mineralization in the control samples revealed a more labile C pool in the subsoil with high respiration rates (Heitkötter et al., 2017a; Wang et al., 2019; Shahzad et al., 2019) at the LS site in both bulk and rhizosphere soil and in the PS rhizosphere supported by relatively higher K_2SO_4 -extractable C (Table 3). This is accompanied by lower metabolic efficiency ($q\text{CO}_2$) in the LS subsoil and thus, higher CO_2 production by microorganisms to meet their energy demands by utilizing more substrate (Lavahun et al., 1996). Further, the metabolic quotient indicates a higher C use efficiency in the subsoil rhizosphere compared to the respective bulk soil showing that microorganism close to roots are adapted to the utilization of low molecular substrates from root exudation (Blagodatskaya et al., 2004; Chen et al., 2016). This shows that despite an overall increasing ^{14}C age with increasing depth (Kögel-Knabner et al., 2008), subsoils can contain a relatively large

proportion of labile SOC as indicated by the relatively higher K_2SO_4 -extractable C especially compared to the topsoil.

4.2. Impact of glucose addition on SOC mineralization and enzyme activities

Contrary to our hypothesis, the addition of ^{14}C labeled glucose led to an inconsistent pattern of SOC mineralization across the three sampling sites. While glucose addition increased the SOC mineralization rates at the LS and TS site it decreased them at the PS site in both bulk and rhizosphere soil. In agreement with our hypothesis, glucose mineralization in the rhizosphere soil exceeded the rates of bulk soil at the beginning of the incubation, most likely caused by faster turnover of this labile substrate by the more abundant and diverse microbial community (Blagodatskaya et al., 2004; Chen et al., 2016). After 5 days of incubation, glucose addition caused a positive priming effect ($P < 0.05$) at the TS site in bulk and rhizosphere soil, indicating that microorganisms have a high energy demand (Guenet et al., 2012) as they rapidly mineralize the added substrate and the existing OM, resulting in an overall elevated SOC mineralization. At the end of the incubation, we still see 33% (bulk) and 44% (rhizosphere) higher mineralization rates accompanied by an increased co-metabolism as indicated by higher DEH activity through microbial growth (Blagodatskaya and Kuzyakov, 2008). Interestingly, glucose addition suppressed the SOC mineralization in the PS bulk and rhizosphere topsoil already after 5 days of incubation (Supplementary Fig. 1) and continued to do so throughout the entire incubation procedure. This might be induced by the preferential utilization of the added substrate instead of mineralizing native OM (Gontikaki et al., 2013). Additionally, this effect is accompanied by suppressed C-cycling enzyme activity (Kuzyakov, 2002; Guenet et al., 2010) when microorganisms only use the added substrate as a C source instead of decomposing the more energy intensive SOC. This apparently was the case at the PS and the TS site since NAC and BGL activity decreased after glucose addition and microorganisms seem to avoid energy intensive enzyme production (Schimel and Weintraub, 2003; Spohn et al., 2016). Generally, enzyme was higher in the rhizosphere soil likely caused by root exudation (Yin et al., 2014; Cheng et al., 2014) providing microorganisms with constant influx of fresh substrate. The addition of ^{14}C labeled glucose in the subsoil showed a similar pattern compared to the topsoil with a rapid substrate mineralization within the first few days of incubation. This fast substrate utilization is accompanied by a positive priming effect at the TS site in both bulk and rhizosphere soil ($P < 0.05$) at day 5. After 82 days of incubation, we still observed an increased SOC mineralization along with co-metabolism as reflected by higher DEH activity in both bulk and rhizosphere soil. Here, the added glucose was efficiently used as an extra energy source to obtain more C from the otherwise stabilized OM compounds. The reduced activity of C-cycling enzymes after glucose addition indicates the presence of labile C in the subsoil (Spohn et al., 2016; Keane et al., 2020) which is supported by the higher relative K_2SO_4 values compared to the topsoil. DEH normalized to SOC content shows that the microorganisms in the investigated subsoil samples seem to be more active than in the topsoil as indicated by the higher relative DEH activity.

4.3. Impact of palmitic acid addition on SOC mineralization and enzyme activities

The addition of ^{14}C labeled palmitic acid increased SOC mineralization at all sites, with the exception of the PS rhizosphere soils. Contrary to the glucose addition, the substrate mineralization of palmitic acid showed no consistent pattern across the three sampling sites with a fast turnover at the TS site and a long retardation period at the PS and LS site. Interestingly, we observed a positive priming effect ($P < 0.05$) already after 5 days of incubation in the PS bulk topsoil. Apparently, the microbial community, which mainly consists of Fungi, Acidobacteria and Actinobacteria in this soil (Herre et al., 2021), seems to be capable

of degrading more complex substrates as also suggested by the high POX activity at this site. After 82 days of incubation, 37% (bulk) and 45% (rhizosphere) of the palmitic acid at the PS site were mineralized after a long retardation period indicating a slow microbial growth and adaption most likely of k-strategist organisms (Fontaine et al., 2003). The palmitic acid treatment at the TS site caused a positive priming effect already at day 5 with 263% (bulk) and 227% (rhizosphere) higher SOC mineralization compared to the respective control sample. We think that the substrate addition stimulated the metabolic capacity of the consumers as the elevated CO₂ production was caused by a simultaneous decomposition of the added substrate and native OM. This could be an indication of a real priming effect even in the early stage of the experiment (Blagodatskaya and Kuzyakov, 2008), which persisted throughout the entire incubation.

In the subsoil, the SOC mineralization was substantially increased in the PS bulk soil by palmitic acid addition already at day 5. However, almost none of the added substrate contributed to this effect as substrate mineralization rate was below 1% from day 1–40 and then slowly increased up to 20%. This effect was also observed by Zyakun et al. (2012) who compared the rates of microbial CO₂ production after n-hexadecanoic acid addition to agricultural and forest soils, and found a long retardation time in the forest samples indicating an adaptation process of the microbial community. At the LS site, the substrate mineralization was also below 1% in both bulk and rhizosphere subsoil samples in the first few days. After day 5 however, the mineralization in the rhizosphere samples quickly increased, indicating a shift in the microbial community composition (Fontaine et al., 2003). Another indication for this adaption to the palmitic acid addition is the percentage of substrate incorporation into the microbial biomass (Supplementary Fig. 2). In all rhizosphere samples, only a small amount of around 2.5% of the added ¹⁴C was incorporated into the biomass while the substrate mineralization was relatively high (around 35%). In the bulk soil however, more ¹⁴C was incorporated at the PS (7.5%) and LS (12%) site, where the substrate mineralization was relatively low. This demonstrates that the microorganisms at these two sites have different energy demands and are using the substrate to build up their biomass. The palmitic acid treatment at the TS site caused a similar reaction as the glucose addition with a fast substrate mineralization already at the beginning of the incubation. This is most likely caused by a different microbial community structure in the subsoil which seems to be adapted to this more complex C source (Goldfarb et al., 2011; Kumar et al., 2015) and is immediately capable of depolymerizing the long chain fatty acids instead of using it to slowly build up their biomass (Supplementary Fig. 2). In another onsite study, Herre et al. (2021) investigated the microbial community composition at this site and found a high abundance of Actinobacteria, β -Proteobacteria and fungi in the TS subsoil which are known as being main decomposers of complex substrates in deeper soil layers (Dashti et al., 2008; Goldfarb et al., 2011) and main sources for POX production (Ragot et al., 2015; Uksa et al., 2017). This is also reflected by the 141-fold and 246-fold higher POX activity in the LS and TS subsoil compared to the respective topsoil samples, indicating that the microbial community especially in the bulk soil is adapted to utilize more complex polymers (Thevenot et al., 2010; Fujii et al., 2013, 2020). Contrary to the topsoil, the substrate addition had a significant impact on enzyme activity as shown by the MANOVA and the sampling site only affected BGL and ARG (Supplementary Table 2). The increased production of BGL and NAC at the PS site in combination with the long lag period of substrate mineralization shows that microorganism invest in enzyme production when the possible energy yield exceed the energy investment (Schimel and Weintraub, 2003; Spohn et al., 2016). This indicates that enzyme production in the subsoil strongly depends on substrate and nutrient availability (Allison and Vitousek, 2005; Hernández and Hobbie, 2010).

4.4. Impact of nutrient additions on SOC mineralization and enzyme activities

The addition of N and the combination of N + P led to a decrease in SOC mineralization in almost all samples with the most pronounced effect in the PS bulk soil, where the SOC mineralization was 90% below the control sample. This suppressed respiration after N-application was also observed in various other studies (Treseder, 2008; Janssens et al., 2010; Chen et al., 2014; Riggs and Hobbie, 2016; Heitkötter et al., 2017b; Wild et al., 2019; Forsmark et al., 2020) and can be attributed to the N-mining theory (Craine et al., 2007). Hence, microorganisms show suppressed activity through increased N availability because no additional energy for N acquisition is required anymore (Moorhead and Sinsabaugh, 2006). This is also reflected in a significant reduction in DEH activity in almost all samples after N addition. The addition of N containing treatments often tend to reduce activity of enzymes associated with the N-cycle (Allison and Vitousek, 2005; Wang et al., 2019) which was also the case in our study. Both ARG and NAC reacted similarly, showing a reduction in activities compared to the respective control sample. These findings affirm the results of Heitkötter et al. (2017b), who investigated enzyme activities at the PS site after N application in the subsoil and found suppressed NAC activity. Without N, the addition of P alone had no effect on SOC mineralization at the LS and TS site but significantly reduced the CO₂ evolution at the PS site in both bulk and rhizosphere soil throughout the entire incubation procedure. This indifferent response to P addition was also observed by Feng and Zhu (2021) in a recent meta-analysis where P only had a minimal priming effect.

The impact of N addition in the subsoil were much more pronounced compared to the topsoil. DEH activity showed significant reduction with a mean reduced activity of 78% in bulk and 70% in rhizosphere soil. N addition also significantly influenced SOC mineralization in the PS rhizosphere and showed great impact at the TS rhizosphere with a reduced mineralization of 90% and 45%, respectively. Interestingly, the SOC mineralization at the LS site did not show any noticeable change in the bulk soil (−2%) and a less pronounced reduction in the rhizosphere soil (−26%) but was higher in average compared to the PS and TS site. This overall high SOC mineralization at the LS site was accompanied by high enzyme activities in the N and N + P treated samples. Our results also support findings by Preusser et al. (2017) and Heitkötter et al. (2017a) who found that microorganisms in the PS subsoil are limited by N. Although N deposition rates in forest ecosystem are generally higher compared to grass land (Bebber, 2021) atmospheric N deposition in Europe and especially northern Germany is declining since 1990 and shows a reduction of −30% in our research area from 2000 to 2015 (Schmitz et al., 2019). It is therefore likely that N limitation will continue to be a challenging factor for microbial life in the investigated soils. With sufficient N supply however, the investigated subsoils seem to be limited by P as PHO activity was increased in the N treated samples resulting in more investment towards P acquiring enzymes to meet the shift in nutrient demand (Vitousek et al., 2010; Marklein and Houlton, 2012; Margalef et al., 2021). At the PS site, P addition had no significant effect in the bulk soil which was also the case in a different incubation study by Heitkötter et al. (2017a) at this site and affirmed by a meta-analysis from Feng and Zhu (2021) who found little to no effect on SOC mineralization after sole P addition. In combination with a C source however, positive priming effects were observed after P addition (Heitkötter et al., 2017a, 2017b; Meyer et al., 2018; Mehnaz et al., 2019). These results indicate that the investigated subsoils are not primarily limited by P but may be co-limited through an imbalance of nutrient content (Fay et al., 2015; Hedwall et al., 2017).

As revealed by the PCA, the variability of the dataset in the topsoil is strongly influenced by sampling type (i.e., bulk and rhizosphere soil) and sampling site with the highest factor loadings from enzyme activity and N content. Enzyme activity in the topsoil was also affected by sampling type and sampling site and only PHO was significantly

influenced by treatment addition (Supplementary Table 2). In the subsoil, the PCA showed that sampling type (i.e., bulk and rhizosphere soil) is a major factor for microbial activity with clear clustering of the data set into these two groups. However, the site effect is much less pronounced and contributes less to the variability of the data set but shows clearly a distinction between parent material as the LS site forms an independent cluster and the PS and TS site are clustered together (Fig. 6).

5. Conclusion

Our research shows that glucose and palmitic acid addition induced different response mechanisms by the microorganisms and resulted in both negative and positive. It became apparent that the sampling site and depth significantly effects microbial activity and that the pre-conditions for potential limitations seem to be controlled by parent material as shown by the higher SOC mineralization at site developed on loess. Our results also show that the rhizosphere soil is a distinct hotspot for microbial life with higher carbon use efficiency and enzyme activity compared to the bulk soil. The significantly different response times to palmitic acid addition between the sample sites also suggests that the microbial community undergoes different adaptation processes based on their unique community composition. This clearly demonstrates that the limitations of microbial activity in forest soils cannot uniformly be predicted and are controlled by factors other than sheer substrate limitation. Our results clearly show substantial N limitation in the investigated soils as reflected by the reduction in SOC mineralization and N acquiring enzyme production after nutrient addition, especially in the Dystric Cambisols. Although hypothesis 1 and 2 have to be partially rejected as substrate and nutrient addition were observed in both sampling types and depths, our study showed faster response time to both high and low molecular C compounds in the rhizosphere soil in both depth indicating that the microbial community in the rhizosphere is more capable of decomposing available substrates.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.soilbio.2022.108775>.

References

- Allison, S.D., Vitousek, P.M., 2005. Responses of extracellular enzymes to simple and complex nutrient inputs. *Soil Biology and Biochemistry* 37, 937–944.
- Angst, G., John, S., Mueller, C.W., Kögel-Knabner, I., Rethemeyer, J., 2016a. Tracing the sources and spatial distribution of organic carbon in subsoils using a multi-biomarker approach. *Scientific Reports* 6, 29478.
- Angst, G., Kögel-Knabner, I., Kirfel, K., Hertel, D., Mueller, C.W., 2016b. Spatial distribution and chemical composition of soil organic matter fractions in rhizosphere and non-rhizosphere soil under European beech (*Fagus sylvatica* L.). *Geoderma* 264, 179–187.
- Bebber, D.P., 2021. The gap between atmospheric nitrogen deposition experiments and reality. *The Science of the Total Environment* 801, 149774.
- Bingeman, C.W., Varner, J.E., Martin, W.P., 1953. The effect of the addition of organic materials on the decomposition of an organic soil. *Soil Science Society of America Journal* 17, 34–38.
- Blagodatskaya, E., Kuzyakov, Y., 2008. Mechanisms of real and apparent priming effects and their dependence on soil microbial biomass and community structure: critical review. *Biology and Fertility of Soils* 45, 115–131.
- Blagodatskaya, E.V., Ermolaev, A.M., Myakshina, T.N., 2004. Ecological strategies of soil microbial communities under plants of meadow ecosystems. *Biology Bulletin* 31, 620–627.
- Bundesanstalt für Bodenforschung, 1973. Geologische Übersichtskarte 1, 200000.
- Bundt, M., Widmer, F., Pesaro, M., Zeyer, J., Blaser, P., 2001. Preferential flow paths: biological ‘hot spots’ in soils. *Soil Biology and Biochemistry* 33, 729–738.
- Burns, R.G., DeForest, J.L., Marxsen, J., Sinsabaugh, R.L., Stromberger, M.E., Wallenstein, M.D., Weintraub, M.N., Zoppini, A., 2013. Soil enzymes in a changing environment: current knowledge and future directions. *Soil Biology and Biochemistry* 58, 216–234.
- Chabbi, A., Kögel-Knabner, I., Rumpel, C., 2009. Stabilised carbon in subsoil horizons is located in spatially distinct parts of the soil profile. *Soil Biology and Biochemistry* 41, 256–261.
- Chen, R., Senbayram, M., Blagodatsky, S., Myachina, O., Dittert, K., Lin, X., Blagodatskaya, E., Kuzyakov, Y., 2014. Soil C and N availability determine the priming effect: microbial N mining and stoichiometric decomposition theories. *Global Change Biology* 20, 2356–2367.
- Chen, Y., Chen, G., Robinson, D., Yang, Z., Guo, J., Xie, J., Fu, S., Zhou, L., Yang, Y., 2016. Large amounts of easily decomposable carbon stored in subtropical forest subsoil are associated with r-strategy-dominated soil microbes. *Soil Biology and Biochemistry* 95, 233–242.
- Cheng, W., Johnson, D.W., Fu, S., 2003. Rhizosphere effects on decomposition: controls of plant species, phenology, and fertilization. *Soil Science Society of America Journal* 67, 1418–1427.
- Cheng, W., Parton, W.J., Gonzalez-Meler, M.A., Phillips, R., Asao, S., McNickle, G.G., Brzostek, E., Jastrow, J.D., 2014. Synthesis and modeling perspectives of rhizosphere priming. *New Phytologist* 201, 31–44.
- Craine, J.M., Morrow, C., Fierer, N., 2007. Microbial nitrogen limitation increases decomposition. *Ecology* 88, 2105–2113.
- Dashti, N., Al-Awadhi, H., Khanafer, M., Abdelghany, S., Radwan, S., 2008. Potential of hexadecane-utilizing soil-microorganisms for growth on hexadecanol, hexadecanal and hexadecanoic acid as sole sources of carbon and energy. *Chemosphere* 70, 475–479.
- Ellenberg, H., Leuschner, C., 2010. Vegetation Mitteleuropas mit den Alpen. In: Botanik, Ökologie, Agrar- und Forstwissenschaften, sixth ed. Ulmer, Stuttgart. Geographie 8104.
- FAO, 2014. World reference base for soil resources 2014: international soil classification system for naming soils and creating legends for soil maps. In: World Soil Resources Reports, vol. 106. FAO, Rome.
- Fay, P.A., Prober, S.M., Harpole, W.S., Knops, J.M.H., Bakker, J.D., Borer, E.T., Lind, E.M., MacDougall, A.S., Seabloom, E.W., Wragg, P.D., Adler, P.B., Blumenthal, D.M., Buckley, Y.M., Chu, C., Cleland, E.E., Collins, S.L., Davies, K.F., Du, G., Feng, X., Firn, J., Gruner, D.S., Hagenah, N., Hautier, Y., Heckman, R.W., Jin, V.L., Kirkman, K.P., Klein, J., Ladwig, L.M., Li, Q., McCulley, R.L., Melbourne, B.A., Mitchell, C.E., Moore, J.L., Morgan, J.W., Risch, A.C., Schütz, M., Stevens, C.J., Wedin, D.A., Yang, L.H., 2015. Grassland productivity limited by multiple nutrients. *Nature Plants* 1, 15080.
- Feng, J., Zhu, B., 2021. Global patterns and associated drivers of priming effect in response to nutrient addition. *Soil Biology and Biochemistry* 153, 108118.
- Feng, J., Tang, M., Zhu, B., 2021. Soil priming effect and its responses to nutrient addition along a tropical forest elevation gradient. *Global Change Biology* 27, 2793–2806.
- Fierer, N., Allen, A.S., Schimel, J.P., Holden, P.A., 2003a. Controls on microbial CO₂ production: a comparison of surface and subsurface soil horizons. *Global Change Biology* 9, 1322–1332.
- Fierer, N., Schimel, J.P., Holden, P.A., 2003b. Variations in microbial community composition through two soil depth profiles. *Soil Biology and Biochemistry* 35, 167–176.
- Fontaine, S., Mariotti, A., Abbadié, L., 2003. The priming effect of organic matter: a question of microbial competition? *Soil Biology and Biochemistry* 35, 837–843.
- Forsmark, B., Nordin, A., Maaroufi, N.I., Lundmark, T., Gundale, M.J., 2020. Low and high nitrogen deposition rates in northern coniferous forests have different impacts on aboveground litter production, soil respiration, and soil carbon stocks. *Ecosystems* 23, 1423–1436.
- Fujii, K., Uemura, M., Hayakawa, C., Funakawa, S., Kosaki, T., 2013. Environmental control of lignin peroxidase, manganese peroxidase, and laccase activities in forest floor layers in humid Asia. *Soil Biology and Biochemistry* 57, 109–115.
- Fujii, K., Nakada, Y., Umezawa, K., Yoshida, M., Shibata, M., Hayakawa, C., Inagaki, Y., Kosaki, T., Hangs, R., 2020. A comparison of lignin-degrading enzyme activities in forest floor layers across a global climatic gradient. *Soil Biology and Biochemistry* 2, 281–294.
- Goldfarb, K.C., Karaoz, U., Hanson, C.A., Santee, C.A., Bradford, M.A., Treseder, K.K., Wallenstein, M.D., Brodie, E.L., 2011. Differential growth responses of soil bacterial taxa to carbon substrates of varying chemical recalcitrance. *Frontiers in Microbiology* 2, 1–10.
- Gontikaki, E., Thornton, B., Huvenne, V.A.I., Witte, U., 2013. Negative priming effect on organic matter mineralisation in NE Atlantic slope sediments. *PLoS One* 8, e67722.

- Guenet, B., Leloup, J., Raynaud, X., Bardoux, G., Abbadié, L., 2010. Negative priming effect on mineralization in a soil free of vegetation for 80 years. *European Journal of Soil Science* 61, 384–391.
- Guenet, B., Juarez, S., Bardoux, G., Abbadié, L., Chenu, C., 2012. Evidence that stable C is as vulnerable to priming effect as is more labile C in soil. *Soil Biology and Biochemistry* 52, 43–48.
- Hamer, U., Marschner, B., 2002. Priming effects of sugars, amino acids, organic acids and catechol on the mineralization of lignin and peat. *Journal of Plant Nutrition and Soil Science* 165, 261–268.
- Hamer, U., Marschner, B., 2005. Priming effects in different soil types induced by fructose, alanine, oxalic acid and catechol additions. *Soil Biology and Biochemistry* 37, 445–454.
- Hedwall, P.-O., Bergh, J., Brunet, J., 2017. Phosphorus and nitrogen co-limitation of forest ground vegetation under elevated anthropogenic nitrogen deposition. *Oecologia* 185, 317–326.
- Heine, S., Ludwig, B., Piepho, H.-P., Mikutta, R., Don, A., Wordell-Dietrich, P., Helfrich, M., Hertel, D., Leuschner, C., Kirfel, K., Kandeler, E., Preusser, S., Guggenberger, G., Leinemann, T., Marschner, B., 2018. Factors controlling the variability of organic matter in the top- and subsoil of a sandy Dystric Cambisol under beech forest. *Geoderma* 311, 37–44.
- Heitkötter, J., Marschner, B., 2018a. Is there anybody out there? Substrate availability controls microbial activity outside of hotspots in subsoils. *Soil Biology and Biochemistry* 2, 35.
- Heitkötter, J., Marschner, B., 2018b. Soil zymography as a powerful tool for exploring hotspots and substrate limitation in undisturbed subsoil. *Soil Biology and Biochemistry* 124, 210–217.
- Heitkötter, J., Heine, S., Marschner, B., 2017a. Relevance of substrate quality and nutrients for microbial C-turnover in top- and subsoil of a Dystric Cambisol. *Geoderma* 302, 89–99.
- Heitkötter, J., Niebuhr, J., Heine, S., Marschner, B., 2017b. Patterns of nitrogen and citric acid induced changes in C-turnover and enzyme activities are different in topsoil and subsoils of a sandy Cambisol. *Geoderma* 292, 111–117.
- Hernández, D.L., Hobbie, S.E., 2010. The effects of substrate composition, quantity, and diversity on microbial activity. *Plant and Soil* 335, 397–411.
- Herre, M., Heitkötter, J., Heine, S., Rethemeyer, J., Preusser, S., Kandeler, E., Marschner, B., 2021. Differences in Organic Matter Properties and Microbial Activity between Bulk and Rhizosphere Soil from the Top- and Subsoils of Three Forest Stands, 115589. *Geoderma*.
- Huo, C., Luo, Y., Cheng, W., 2017. Rhizosphere priming effect: a meta-analysis. *Soil Biology and Biochemistry* 111, 78–84.
- Janssens, I.A., Dieleman, W., Luyssaert, S., Subke, J.-A., Reichstein, M., Ceulemans, R., Ciais, P., Dolman, A.J., Grace, J., Matteucci, G., Papale, D., Piao, S.L., Schulze, E.-D., Tang, J., Law, B.E., 2010. Reduction of forest soil respiration in response to nitrogen deposition. *Nature Geoscience* 3, 315–322.
- Jenkinson, D.S., 1966. The priming action. In: Report of the FAO/IAEA Technical Meeting Organized by the Food and Agriculture Organization of the United Nations and the International Atomic Energy Agency in Cooperation with the International Soil Society Brunswick-Volkenrode. Pergamon Press, pp. 199–208, 9–14 September 1963.
- Jenkinson, D.S., Poulton, P.R., Bryant, C., 2008. The turnover of organic carbon in subsoils. Part 1. Natural and bomb radiocarbon in soil profiles from the Rothamsted long-term field experiments. *European Journal of Soil Science* 59, 391–399.
- Jia, J., Cao, Z., Liu, C., Zhang, Z., Lin, L., Wang, Y., Haghpor, N., Wacker, L., Bao, H., Dittmar, T., Simpson, M.J., Yang, H., Crowther, T.W., Eglinton, T.I., He, J.-S., Feng, X., 2019. Climate warming alters subsoil but not topsoil carbon dynamics in alpine grassland. *Global Change Biology* 25, 4383–4393.
- Jobbagy, E.G., Jackson, R.B., 2000. The vertical distribution of soil organic carbon and its relation to climate and vegetation. *Ecological Applications* 10, 423.
- Joergensen, R.G., Wichern, F., 2018. Alive and kicking: why dormant soil microorganisms matter. *Soil Biology and Biochemistry* 116, 419–430.
- Keane, J.B., Hoosbeek, M.R., Taylor, C.R., Miglietta, F., Phoenix, G.K., Hartley, I.P., 2020. Soil C, N and P cycling enzyme responses to nutrient limitation under elevated CO₂. *Biogeochemistry* 151, 221–235.
- Kirfel, K., Heine, S., Hertel, D., Leuschner, C., 2019. Effects of bedrock type and soil chemistry on the fine roots of European beech – a study on the belowground plasticity of trees. *Forest Ecology and Management* 444, 256–268.
- Kögel-Knabner, I., Guggenberger, G., Kleber, M., Kandeler, E., Kalbitz, K., Scheu, S., Eusterhues, K., Leinweber, P., 2008. Organo-mineral associations in temperate soils: integrating biology, mineralogy, and organic matter chemistry. *Journal of Plant Nutrition and Soil Science* 171, 61–82.
- Kumar, M., Singh, J., Singh, M.K., Singhal, A., Thakur, I.S., 2015. Investigating the degradation process of kraft lignin by β -proteobacterium, *Pandoraea* sp. ISTKB. *Environmental Science and Pollution Research* 22, 15690–15702.
- Kuzyakov, Y., 2002. Review: factors affecting rhizosphere priming effects. *Journal of Plant Nutrition and Soil Science* 165, 382–396.
- Kuzyakov, Y., 2010. Priming effects: interactions between living and dead organic matter. *Soil Biology and Biochemistry* 42, 1363–1371.
- Kuzyakov, Y., Blagodatskaya, E., 2015. Microbial hotspots and hot moments in soil: concept & review. *Soil Biology and Biochemistry* 83, 184–199.
- Kuzyakov, Y., Friedel, J., Stahr, K., 2000. Review of mechanisms and quantification of priming effects. *Soil Biology and Biochemistry* 32, 1485–1498.
- LaRowe, D.E., van Cappellen, P., 2011. Degradation of natural organic matter: a thermodynamic analysis. *Geochimica et Cosmochimica Acta* 75, 2030–2042.
- Lavahun, M.F.E., Joergensen, R.G., Meyer, B., 1996. Activity and biomass of soil microorganisms at different depths. *Biology and Fertility of Soils* 23, 38–42.
- Liu, H., Mi, Z., Lin, L., Wang, Y., Zhang, Z., Zhang, F., Wang, H., Liu, L., Zhu, B., Cao, G., Zhao, X., Sanders, N.J., Classen, A.T., Reich, P.B., He, J.-S., 2018. Shifting plant species composition in response to climate change stabilizes grassland primary production. *Proceedings of the National Academy of Sciences* 115, 4051–4056.
- Margalef, O., Sardans, J., Maspons, J., Molowny-Horas, R., Fernández-Martínez, M., Janssens, I.A., Richter, A., Ciais, P., Obersteiner, M., Peñuelas, J., 2021. The effect of global change on soil phosphatase activity. *Global Change Biology* 27, 5989–6003.
- Marklein, A.R., Houlton, B.Z., 2012. Nitrogen inputs accelerate phosphorus cycling rates across a wide variety of terrestrial ecosystems. *New Phytologist* 193, 696–704.
- Marx, M.-C., Wood, M., Jarvis, S., 2001. A microplate fluorimetric assay for the study of enzyme diversity in soils. *Soil Biology and Biochemistry* 33, 1633–1640.
- Mehnaz, K.R., Corneo, P.E., Keitel, C., Dijkstra, F.A., 2019. Carbon and phosphorus addition effects on microbial carbon use efficiency, soil organic matter priming, gross nitrogen mineralization and nitrous oxide emission from soil. *Soil Biology and Biochemistry* 134, 175–186.
- Mersi, W.von, Schinner, F., 1991. An improved and accurate method for determining the dehydrogenase activity of soils with iodinitroretetrazolium chloride. *Biology and Fertility of Soils* 11, 216–220.
- Meyer, N., Welp, G., Rodionov, A., Borchard, N., Martius, C., Amelung, W., 2018. Nitrogen and phosphorus supply controls soil organic carbon mineralization in tropical topsoil and subsoil. *Soil Biology and Biochemistry* 119, 152–161.
- Moorhead, D.L., Sinsabaugh, R.L., 2006. A theoretical model of litter decay and microbial interaction. *Ecological Monographs* 76, 151–174.
- O'Rourke, S.M., Angers, D.A., Holden, N.M., McBratney, A.B., 2015. Soil organic carbon across scales. *Global Change Biology* 21, 3561–3574.
- Preusser, S., Marhan, S., Poll, C., Kandeler, E., 2017. Microbial community response to changes in substrate availability and habitat conditions in a reciprocal subsoil transfer experiment. *Soil Biology and Biochemistry* 105, 138–152.
- Ragot, S.A., Kertész, M.A., Bünnemann, E.K., 2015. phoD alkaline phosphatase gene diversity in soil. *Applied and Environmental Microbiology* 81, 7281–7289.
- Rasse, D.P., Rumpel, C., Dignac, M.-F., 2005. Is soil carbon mostly root carbon? Mechanisms for a specific stabilisation. *Plant and Soil* 269, 341–356.
- Rethemeyer, J., Kramer, C., Gleixner, G., John, B., Yamashita, T., Flessa, H., Andersen, N., Nadeau, M.-J., Grootes, P.M., 2005. Transformation of organic matter in agricultural soils: radiocarbon concentration versus soil depth. *Geoderma* 128, 94–105.
- Riggs, C.E., Hobbie, S.E., 2016. Mechanisms driving the soil organic matter decomposition response to nitrogen enrichment in grassland soils. *Soil Biology and Biochemistry* 99, 54–65.
- Rumpel, C., Kögel-Knabner, I., 2011. Deep soil organic matter—a key but poorly understood component of terrestrial C cycle. *Plant and Soil* 338, 143–158.
- Salaomé, C., Nunan, N., Pouteau, V., Lerch, T.Z., Chenu, C., 2010. Carbon dynamics in topsoil and in subsoil may be controlled by different regulatory mechanisms. *Global Change Biology* 16, 416–426.
- Schimel, J., Weintraub, M.N., 2003. The implications of exoenzyme activity on microbial carbon and nitrogen limitation in soil: a theoretical model. *Soil Biology and Biochemistry* 35, 549–563.
- Schimel, J., Balser, T.C., Wallenstein, M., 2007. Microbial stress-response physiology and its implications for ecosystem function. *Ecology* 88, 1386–1394.
- Schmitz, A., Sanders, T.G.M., Bolte, A., Bussotti, F., Dirnböck, T., Johnson, J., Peñuelas, J., Pollastrini, M., Prescher, A.-K., Sardans, J., Verstraeten, A., Vries, W.de, 2019. Responses of forest ecosystems in Europe to decreasing nitrogen deposition. *Environmental Pollution* 244, 980–994.
- Schöning, I., Kögel-Knabner, I., 2006. Chemical composition of young and old carbon pools throughout Cambisol and Luvisol profiles under forests. *Soil Biology and Biochemistry* 38, 2411–2424.
- Schöning, I., Totsche, K.U., Kögel-Knabner, I., 2006. Small scale spatial variability of organic carbon stocks in litter and solum of a forested Luvisol. *Geoderma* 136, 631–642.
- Shahzad, T., Anwar, F., Hussain, S., Mahmood, F., Arif, M.S., Sahar, A., Nawaz, M.F., Pervene, N., Sanaullah, M., Rehman, K., Rashid, M.I., 2019. Carbon dynamics in surface and deep soil in response to increasing litter addition rates in an agro-ecosystem. *Geoderma* 333, 1–9.
- Sinsabaugh, R.L., 2010. Phenol oxidase, peroxidase and organic matter dynamics of soil. *Soil Biology and Biochemistry* 42, 391–404.
- Sinsabaugh, R., Saiya-Cork, K., Long, T., Osgood, M., Neher, D., Zak, D., Norby, R., 2003. Soil microbial activity in a Liquidambar plantation unresponsive to CO₂-driven increases in primary production. *Applied Soil Ecology* 24, 263–271.
- Soong, J.L., Castanha, C., Hicks Pries, C.E., Ofiti, N., Porras, R.C., Riley, W.J., Schmidt, M.W.I., Torn, M.S., 2021. Five years of whole-soil warming led to loss of subsoil carbon stocks and increased CO₂ efflux. *Science Advances* 7, eabd1343.
- Spohn, M., Klaus, K., Wanek, W., Richter, A., 2016. Microbial carbon use efficiency and biomass turnover times depending on soil depth – implications for carbon cycling. *Soil Biology and Biochemistry* 96, 74–81.
- Thevenot, M., Dignac, M.-F., Rumpel, C., 2010. Fate of lignins in soils: a review. *Soil Biology and Biochemistry* 42, 1200–1211.
- Treseder, K.K., 2008. Nitrogen additions and microbial biomass: a meta-analysis of ecosystem studies. *Ecology Letters* 11, 1111–1120.
- Uksa, M., Buegger, F., Gschwendtner, S., Lueders, T., Kublik, S., Kautz, T., Athmann, M., Köpke, U., Munch, J.C., Schloter, M., Fischer, D., 2017. Bacteria utilizing plant-derived carbon in the rhizosphere of *Triticum aestivum* change in different depths of an arable soil. *Environmental Microbiology Reports* 9, 729–741.
- Vance, E.D., Brookes, P.C., Jenkinson, D.S., 1987. An extraction method for measuring soil microbial biomass C. *Soil Biology and Biochemistry* 19, 703–707.

- Vitousek, P.M., Porder, S., Houlton, B.Z., Chadwick, O.A., 2010. Terrestrial phosphorus limitation: mechanisms, implications, and nitrogen-phosphorus interactions. *Ecological Applications* 20, 5–15.
- Wang, K., Peng, C., Zhu, Q., Zhou, X., Wang, M., Zhang, K., Wang, G., 2017a. Modeling global soil carbon and soil microbial carbon by integrating microbial processes into the ecosystem process model TRIPLEX-GHG. *Journal of Advances in Modeling Earth Systems* 9, 2368–2384.
- Wang, P., Limpens, J., Mommer, L., Ruijven, J., Nauta, A.L., Berendse, F., Schaepman-Strub, G., Blok, D., Maximov, T.C., Heijmans, M.M., 2017b. Above- and below-ground responses of four tundra plant functional types to deep soil heating and surface soil fertilization. *Journal of Ecology* 105, 947–957.
- Wang, D., Zhu, Z., Shahbaz, M., Chen, L., Liu, S., Inubushi, K., Wu, J., Ge, T., 2019. Split N and P addition decreases straw mineralization and the priming effect of a paddy soil: a 100-day incubation experiment. *Biology and Fertility of Soils* 55, 701–712.
- Wild, B., Li, J., Pihlblad, J., Bengtson, P., Rütting, T., 2019. Decoupling of priming and microbial N mining during a short-term soil incubation. *Soil Biology and Biochemistry* 129, 71–79.
- Wolinska, A., Stepniewski, Z., 2012. In: Canuto, R.A. (Ed.), *Dehydrogenase Activity in the Soil Environment, Dehydrogenases*. InTech, pp. 183–210.
- Wordell-Dietrich, P., Don, A., Helfrich, M., 2017. Controlling factors for the stability of subsoil carbon in a Dystric Cambisol. *Geoderma* 304, 40–48.
- Wu, J., Joergensen, R.G., Pommerening, B., Chaussod, R., Brookes, P.C., 1990. Measurement of soil microbial biomass C by fumigation-extraction—an automated procedure. *Soil Biology and Biochemistry* 22, 1167–1169.
- Yanai, R.D., Majdi, H., Park, B.B., 2003. Measured and modelled differences in nutrient concentrations between rhizosphere and bulk soil in a Norway spruce stand. *Plant and Soil* 257, 133–142.
- Yin, H., Wheeler, E., Phillips, R.P., 2014. Root-induced changes in nutrient cycling in forests depend on exudation rates. *Soil Biology and Biochemistry* 78, 213–221.
- Zhu, B., Gutknecht, J.L., Herman, D.J., Keck, D.C., Firestone, M.K., Cheng, W., 2014. Rhizosphere priming effects on soil carbon and nitrogen mineralization. *Soil Biology and Biochemistry* 76, 183–192.
- Zyakun, A., Nii-Annang, S., Franke, G., Fischer, T., Buegger, F., Dilly, O., 2012. Microbial activity and $^{13}\text{C}/^{12}\text{C}$ ratio as evidence of N-hexadecane and N-hexadecanoic acid biodegradation in agricultural and forest soils. *Geomicrobiology Journal* 29, 570–584.